

MOUNTAINS OF WASTE...

THE GLOBAL CHALLENGE OF TRASH

*An Analysis
of Country-Level Data
on Municipal Solid Waste*

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HUMANS GENERATE

2.01 BILLION TONS

OF WASTE ANNUALLY...

OVER 33%

**OF GLOBAL
WASTE IS
MISMANAGED**



The background of the image is a large, sprawling pile of waste and debris, including plastic, metal, and organic matter, under a grey, overcast sky. The text is overlaid on this scene.

WHAT DRIVES WASTE
GENERATION?

POPULATION? WEALTH?
MISMANAGEMENT?

OUTLINE

1. PROBLEM
2. DATA
3. METHODS
4. RESULTS
5. CONCLUSION

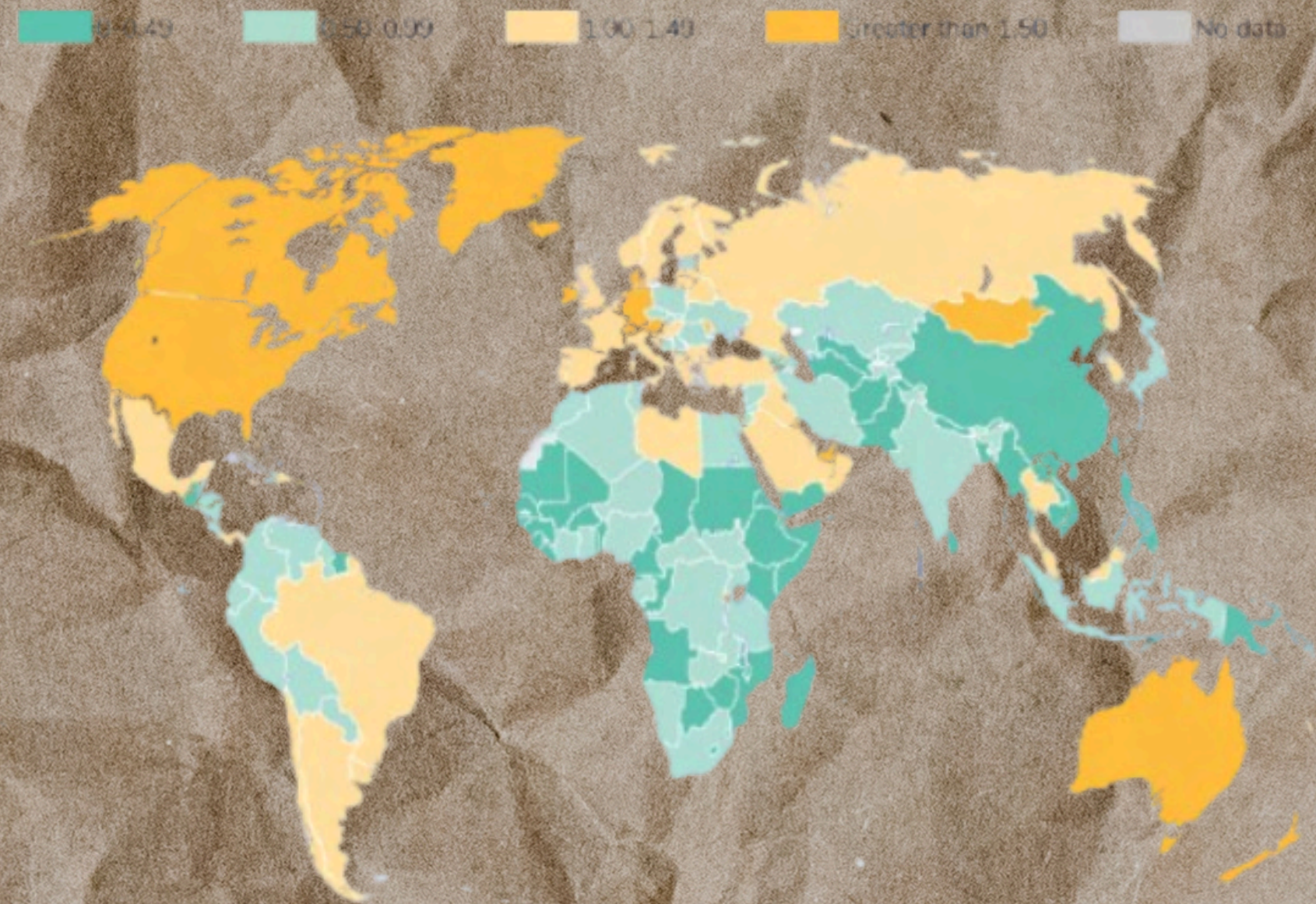
THE PROBLEM GLOBAL WASTE CRISIS



*Waste generation is
outpacing population and economic
growth, creating a global
environmental emergency.*

Impacts:

- *Exacerbation of climate change*
- *Pollution of ecosystems*



DATA

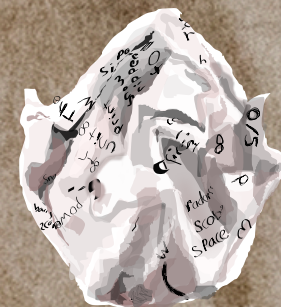
WHAT A WASTE PROJECT

THE WORLD BANK



CSV FILE

"COUNTRY_LEVEL_DATA"



10

COLUMNS

COUNTRY
NAME, GDP,
POPULATION,
TOTAL MSW,
& WASTE
COMPOSITION.

217

ROWS

COUNTRIES

METHODS

VISUALIZATION

- PIE CHART
- SCATTERPLOTS
- CORRELATION MATRIX HEAT MAP



DATA CLEANING

TO ADDRESS
MISSING DATA

USER INTERACTION

FOR COMPARISON

RESULTS

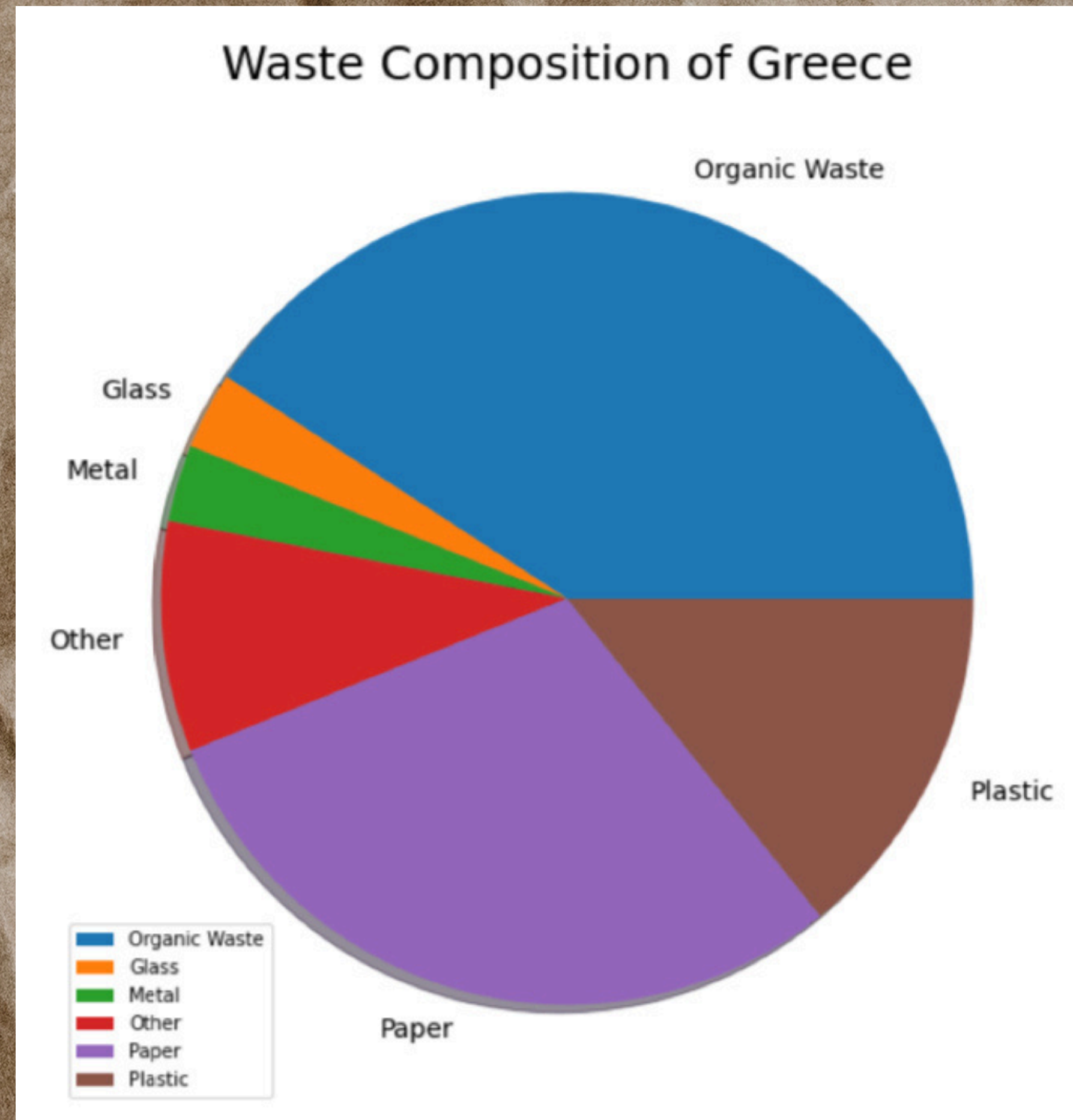
OUTPUT WINDOW

```
Enter country name: greece
Greece GDP world rank (lowest to highest): 152
Greece Waste Generation rank (lowest to highest): 166
Approx. how much waste does your household generate every year? (tons)
10
Approx. what percentage of the waste do you recycle?
5
You have an above-average waste generation! RECYCLE!
You contribute 0.00017 % of the Municipal Solid Waste generated by Greece
Correlation between GDP and MSW: 0.041312873873372666
Correlation between GDP and Population: -0.061642058758295495
Correlation between Population and MSW: 0.8796146115767257
```

RESULTS

PIE CHART

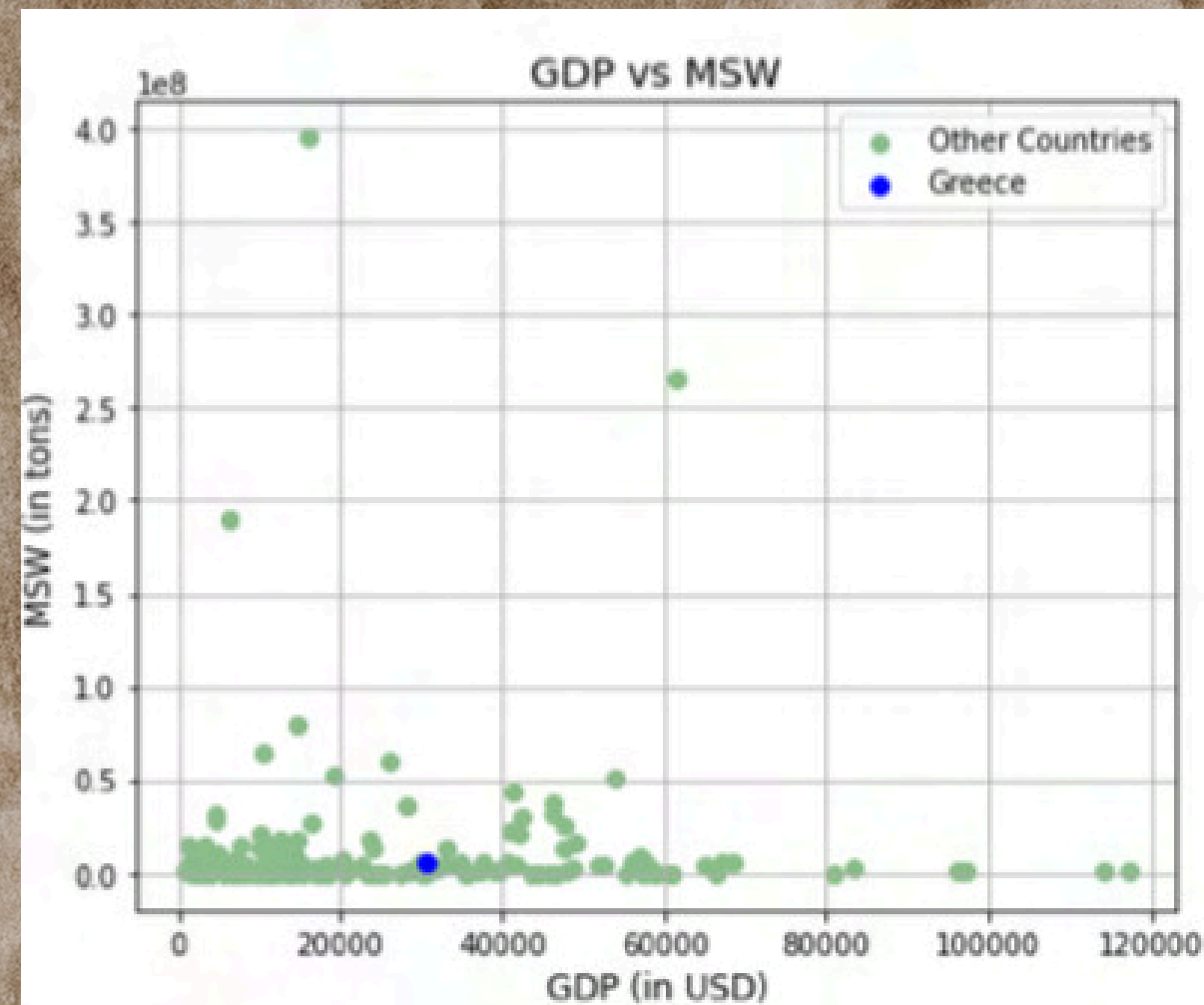
WASTE COMPOSITION OF GREECE



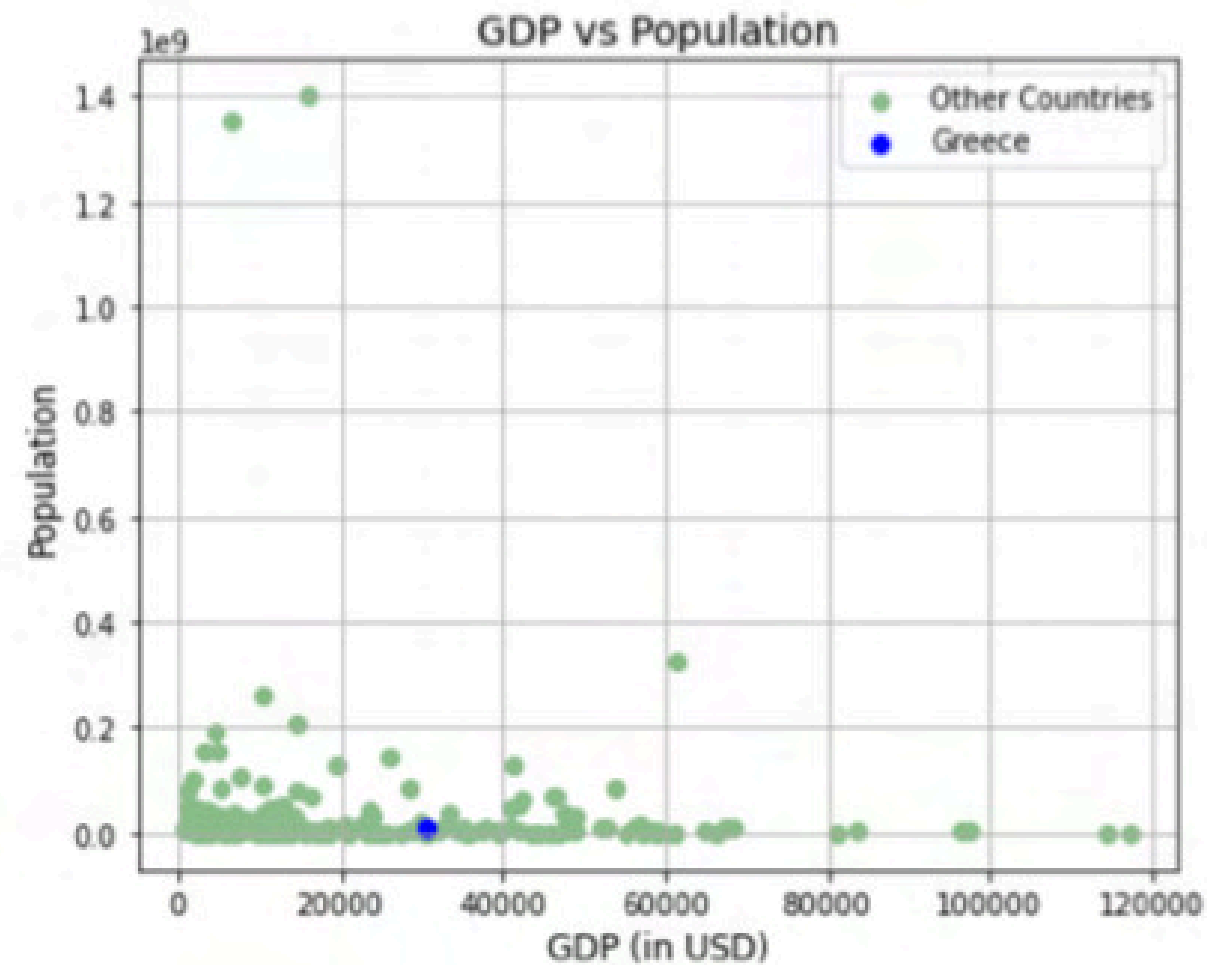
RESULTS

SCATTERPLOTS

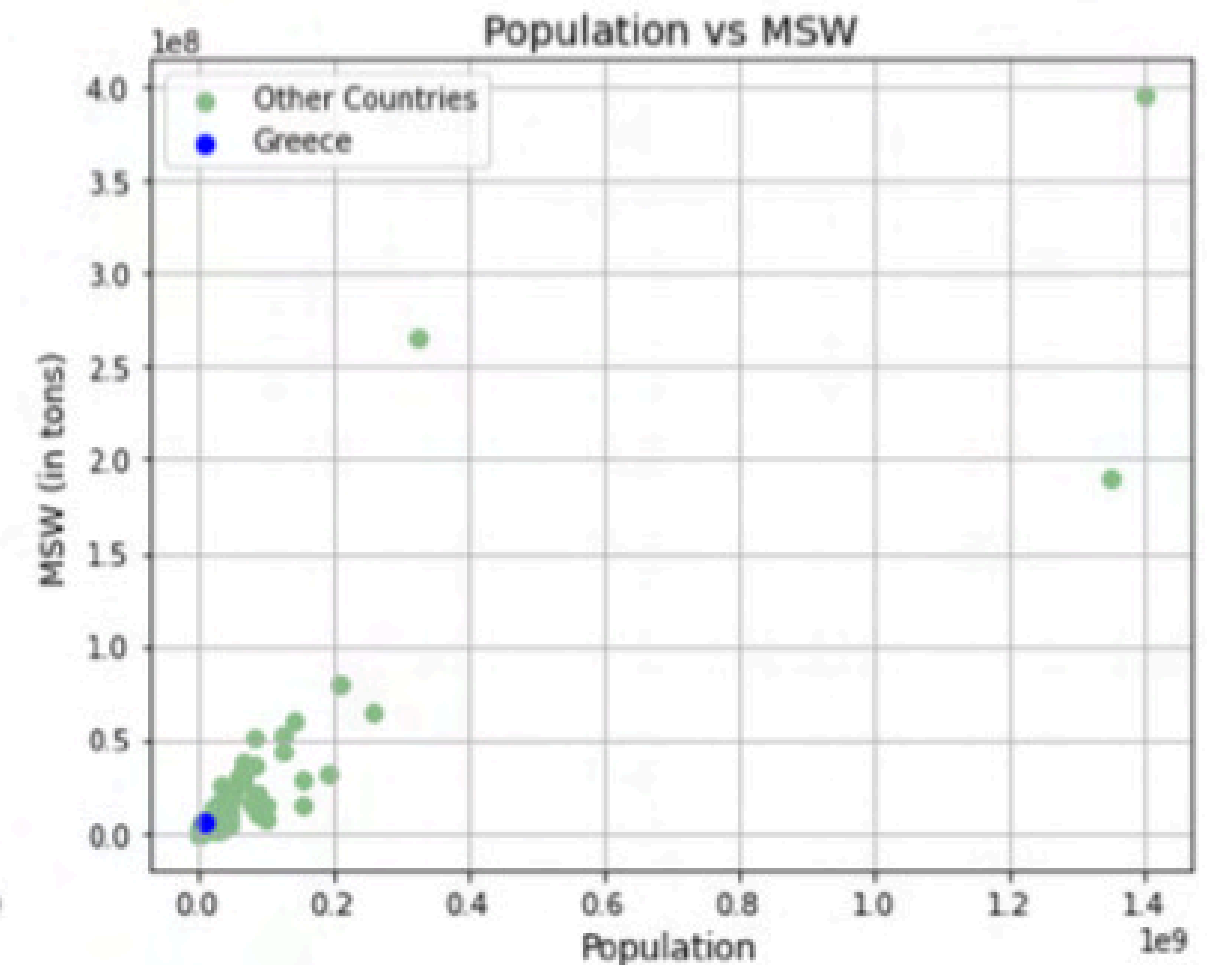
GDP VS MSW
NO STRONG TREND



GDP VS POPULATION
NO STRONG TREND



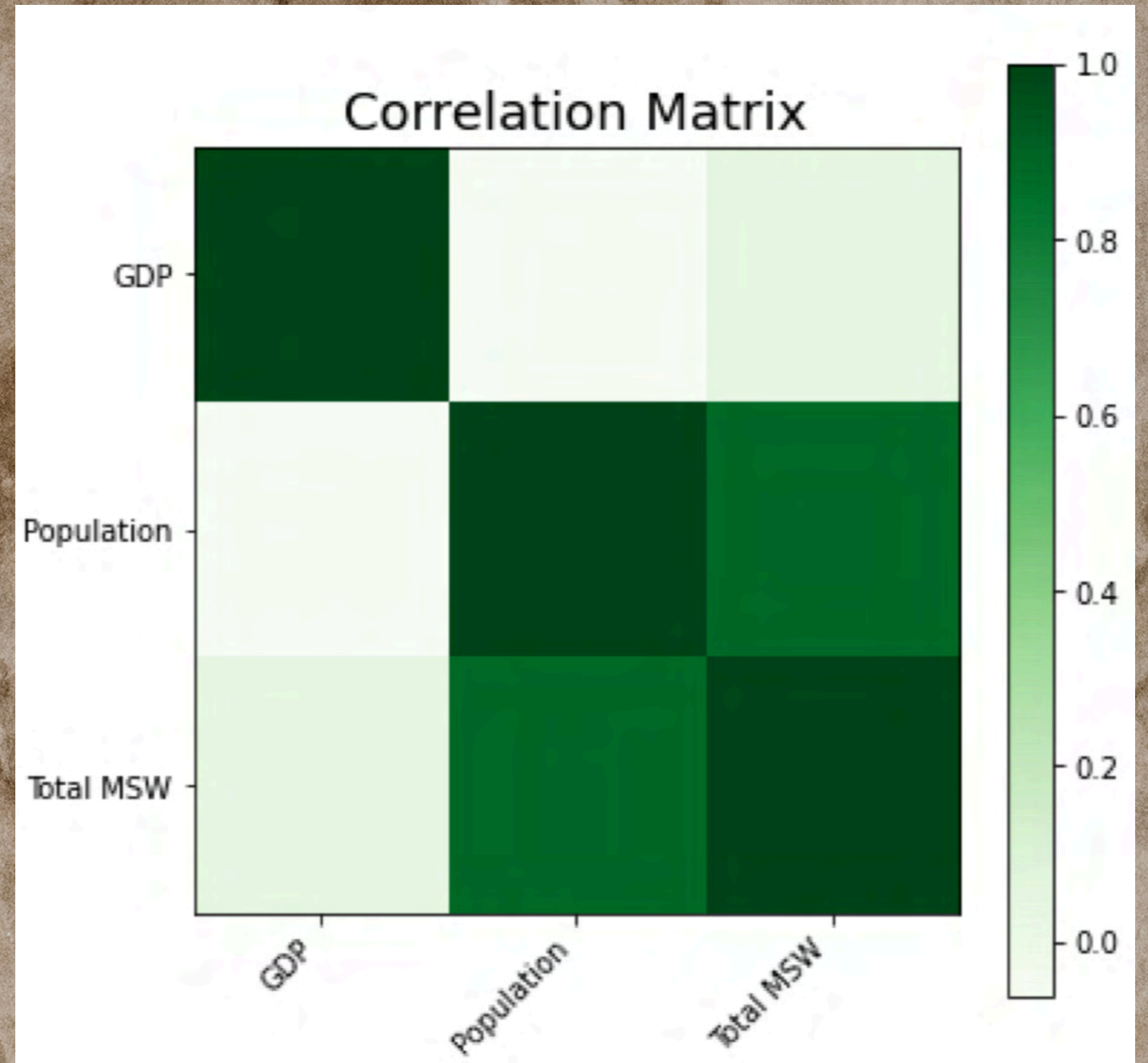
POPULATION VS MSW
POSITIVE RELATIONSHIP



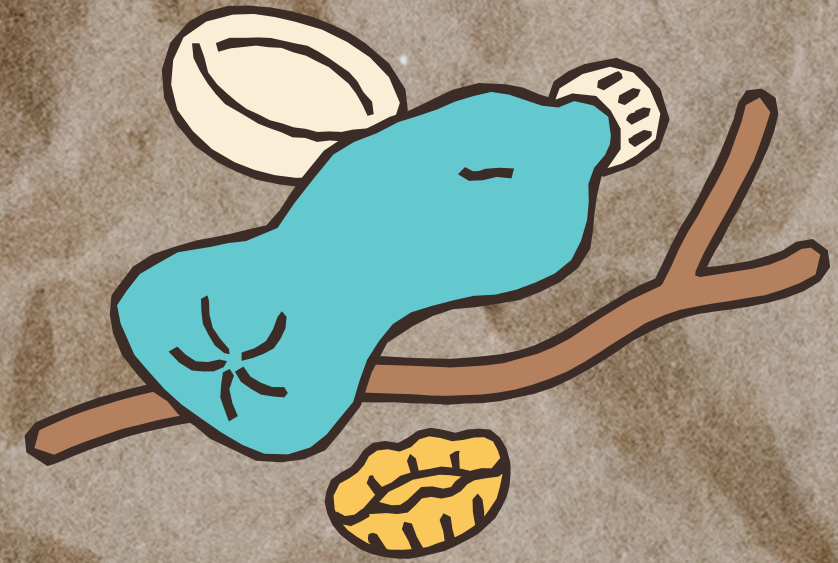
RESULTS

CORRELATION MATRIX

HEATMAP OF
CORRELATIONS BETWEEN
GDP, MSW AND
POPULATIONS



CONCLUSION



POPULATION GROWTH

IS A STRONGER DRIVER OF WASTE GENERATION THAN GDP.

IMPLICATIONS?

FOCUS ON HIGH-POPULATION REGIONS
FOR WASTE MANAGEMENT STRATEGIES.

- IMPROVE PERSONAL WASTE HABITS.
- ADVOCATE FOR RECYCLING POLICIES AND COMPOSTING INITIATIVES.

FUTURE WORK

TIME-SERIES DATA

TO TRACK TRENDS AND PREDICT FUTURE GROWTH

IMPACT OF STRATEGIES

THE IMPACT OF RECYCLING RATES AND GOVERNMENT POLICIES.





THANKS FOR LISTENING!

ACKNOWLEDGEMENTS:

- World Bank's What a Waste Project.
- Support from instructors and peers.

REFERENCES:

The World Bank. "What a Waste."

Datatopics.worldbank.org, 2018,
datatopics.worldbank.org/what-a-waste/.

Blagoeva, Nadezhda, et al. "Relationship between GDP and Municipal Waste; Sustainability, vol. 15, no. 21, 1 Jan. 2023, p. 15193, <https://doi.org/10.3390/su152115193>.